KPP Front-Speed Experiments (Sections 4.3 and 4.4)

KPP reproduction group

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This excerpt reports the Kolmogorov–Petrovsky–Piskunov (KPP) experiments from Shen et al. (arXiv:2601.20024v1). Section 4.3 treats a two-dimensional cellular flow on $\mathbb{T}^2$. Section 4.4 treats a three-dimensional time-dependent Kolmogorov flow on $\mathbb{T}^3$. The two-stage transport map is defined earlier in the group manuscript: a one-step Meanflow map, followed by a near-identity Deep Particle (DP) corrector. The numbers and figures below are from the current reproduction, not from the source paper. Where the source paper is cited, both values are named.

4.3 KPP Front Speed Experiment on the Two-Torus

4.3.1 Setting and protocol

Front propagation in periodic flows supplies a scalar check on the learned maps. The KPP reaction–diffusion–advection equation is

$$u_t + v(x) \cdot \nabla u = \kappa \Delta u + ru(1-u), \tag{1}$$

where κ is the diffusion constant, v is a prescribed velocity field, and u is the reactant concentration. When v is periodic in space and time, the minimal front speed along a direction e has the variational form

$$c^*(e) = \inf_{\lambda > 0} \frac{\mu(\lambda)}{\lambda}, \tag{2}$$

where $\mu(\lambda)$ is the principal eigenvalue of a twisted parabolic operator. For fixed λ, the associated twisted diffusion on $\mathbb{T}^2 = [0, 2\pi)^2$ is

$$dX_t = \big(2\kappa\lambda e_1 + v(X_t, t)\big)\, dt + \sqrt{2\kappa}\, dW_t. \tag{3}$$

The cellular velocity field is

$$v(x,t) = \big(-\cos x_2 - \theta \cos(2\pi t) \sin x_1,\ \cos x_1 + \theta \cos(2\pi t) \sin x_2\big), \tag{4}$$

and the Feynman–Kac potential is

$$V_\lambda(x,t) = \kappa\lambda^2 + \lambda v(x,t) \cdot e_1. \tag{5}$$

The reference clouds are generated by a genetic interacting particle method [1]. The first generation is drawn uniformly on $\mathbb{T}^2$. Each later generation advances one forcing period by Euler–Maruyama steps, multiplies each weight by $\exp(V_\lambda\, dt)$, and resamples once per period. After a long horizon the terminal positions are stored as an empirical sample of the invariant measure $\eta^\star_\lambda$. The running estimators are

$$\hat{\mu}_T(\lambda) = \kappa + \frac{1}{T}\sum_{t=1}^{Tr_n} \log \bar{w}_t, \qquad \hat{c}_T(\lambda) = \frac{\hat{\mu}_T(\lambda)}{\lambda}. \tag{6}$$

The present clouds use $(\lambda, \theta) = (2.0, 1.0)$, $r_n = 256$ steps per period ($dt = 2^{-8}$), $M = 2 \times 10^4$ particles, and $T = 4096$ generations. One terminal cloud is stored per σ and is the target measure for Meanflow, DP, and the histogram W_2. Training and evaluation reuse those stored clouds and do not rerun the long-horizon particle system. Here $\sigma = \kappa$. Training uses $\log_2 \sigma = -2.00, -2.25, \ldots, -3.75$; evaluation extends to $\log_2 \sigma = -4.50$, so the last three levels are extrapolations. Warm starts use the one-step Meanflow map from a Gaussian source with standard deviation near π, optionally composed with a six-block residual corrector on the torus. Both maps are conditioned on σ. Two metrics are recorded: $\hat{c}_T(\lambda)$ at dyadic generations $T = 1, 2, 4, \ldots$, and the torus W_2 on a 72×72 histogram with the periodic ground metric. The uniform first generation of the particle method is the cold start, not the Meanflow source.

4.3.2 Invariant measure

The Feynman–Kac reference concentrates on an anisotropic ridge aligned with the cellular flow. As σ decreases, the ridge tightens, consistent with weaker Brownian spreading in the twisted diffusion. Figure 5 compares that reference with Meanflow and DP at a training level, $\sigma = 2^{-3.75}$, and at an extrapolation level, $\sigma = 2^{-4}$. Meanflow follows the same wavy trough as a continuous band, rather than collapsing to a single filament. The generated ridge is smoother than the Feynman–Kac grain and tightens at the smaller σ. DP stays on that band and lowers the histogram error at both $\sigma = 2^{-3.75}$ and $\sigma = 2^{-4}$. The remaining gap is geometric, not a failure to leave the uniform measure.

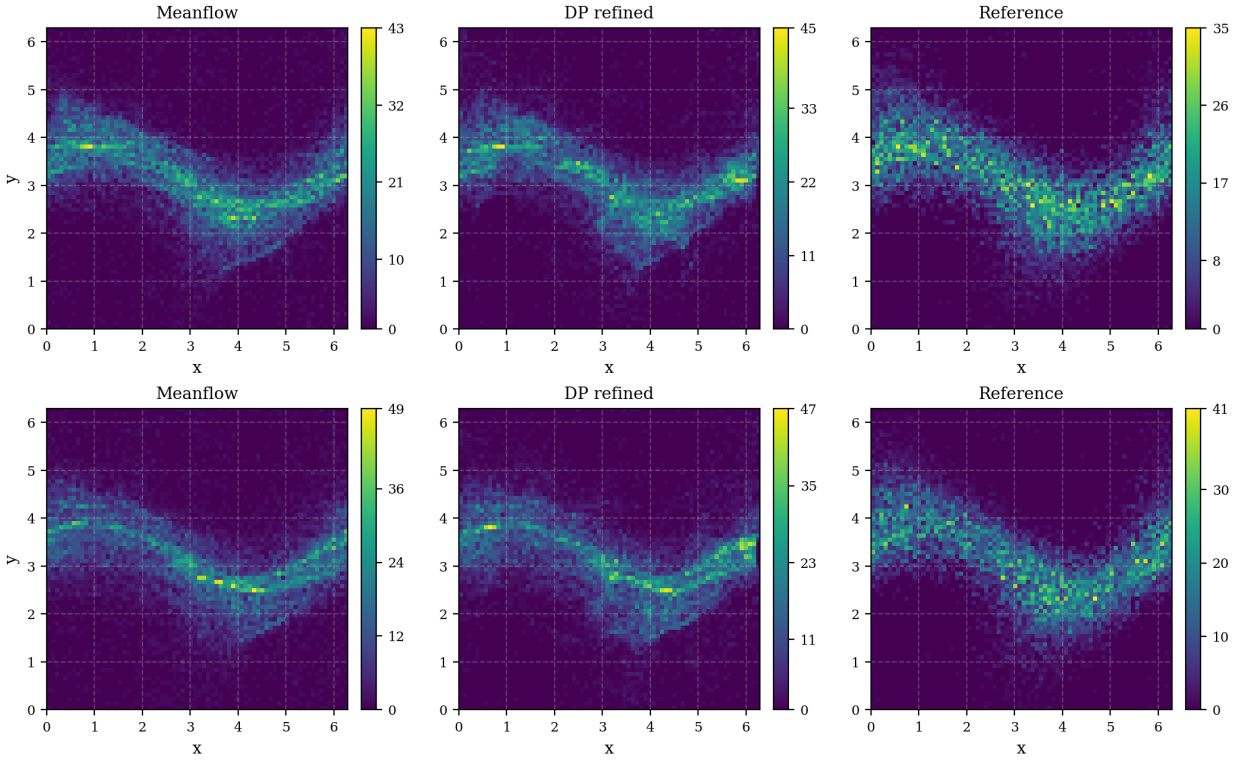


Figure 5: Empirical invariant measures on $\mathbb{T}^2$. Rows are $\sigma = 2^{-3.75}$ (training range) and $\sigma = 2^{-4}$ (extrapolation). Columns are Meanflow, DP, and the long-horizon Feynman–Kac reference. Each panel is a 72×72 count histogram.

4.3.3 Torus Wasserstein Error

A uniform cold start remains between $W_2 = 1.0538$ and 1.1718 on the eleven-level grid. Meanflow lowers the error to a grid mean of 0.1434, about 1.9 times the source-paper Meanflow mean of 0.0769. DP has a grid mean of 0.1345 and is below Meanflow on eight of the eleven levels, including all three extrapolations. At $-\log_2 \sigma = 2.00$ the two values are 0.0962 and 0.1061. At the end of the training range, $-\log_2 \sigma = 3.75$, Meanflow is 0.1424 and DP is 0.1368. At the farthest extrapolation point, $-\log_2 \sigma = 4.50$, Meanflow is 0.2629 and DP is 0.2326, against 0.0456 in the source paper. Figure 6 plots these two columns. The green line marks the end of the training range. Meanflow error grows past that boundary. The corrector lowers the error on the finer training levels and on the extrapolation segment, and does not close the gap to the source-paper DP column.

$-\log_2 \sigma$	Meanflow	DP refinement
2.00	0.0962	0.1061
2.25	0.0962	0.1023
2.50	0.0965	0.1019
2.75	0.1203	0.1138
3.00	0.1075	0.0983
3.25	0.1207	0.1074
3.50	0.1259	0.1123
3.75	0.1424	0.1368
4.00	0.1869	0.1701
4.25	0.2220	0.1978
4.50	0.2629	0.2326

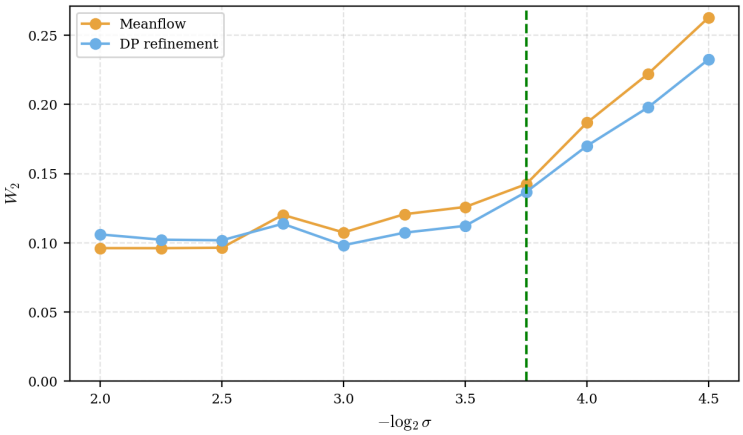


Figure 6: Two-dimensional torus W_2 between the generated cloud and the Feynman–Kac reference. The left table and the right curves use the same Meanflow and DP values. The dashed vertical line is $-\log_2 \sigma = 3.75$.

4.3.4 Eigenvalue estimator

At $\sigma = 2^{-4.5}$, the long-horizon Feynman–Kac value is $\hat{c} = 0.7842$. Figure 7 shows that a uniform cold start begins near 0.51 and reaches the reference after about 10^2 generations. The Meanflow warm start is 0.8180 at $T = 1$ and 0.7841 at $T = 128$. The DP warm start is 0.8167 at $T = 1$ and 0.7848 at $T = 128$. The source paper's DP warm start is 0.7616 at $T = 1$. The log-weight estimator is less sensitive to ridge shape than the histogram W_2, so a warm start can sit near the Feynman–Kac speed while the invariant-measure error remains of order 10^{-1}.

iteration	MF-only init	MF+DP init
1	0.818047	0.816743
2	0.801767	0.798617
4	0.779416	0.780315
8	0.783682	0.783460
16	0.783696	0.782235
32	0.784882	0.784091
64	0.783953	0.784331
128	0.784138	0.784818

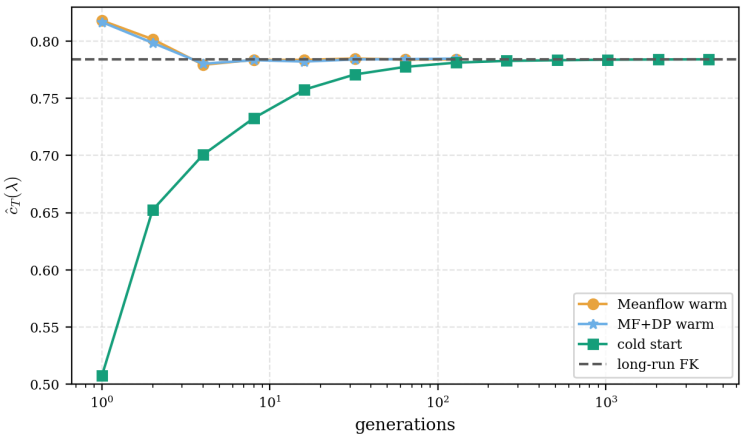


Figure 7: Two-dimensional front-speed estimator $\hat{c}_T(\lambda)$ at $\sigma = 2^{-4.5}$. The left table lists the two warm starts. The right panel adds the uniform cold start and the long-horizon Feynman–Kac reference (dashed). The horizontal axis is logarithmic in the number of generations.

4.4 3D KPP Front Speed Experiment with Time-Dependent Kolmogorov Flow

4.4.1 Setting and protocol

The same KPP equation is posed on $\mathbb{T}^3 = [0, 2\pi)^3$, with the time-dependent Kolmogorov field

$$v(x,t) = \big(\sin(x_3 + \sin(2\pi t)),\ \sin(x_1 + \sin(2\pi t)),\ \sin(x_2 + \sin(2\pi t))\big). \tag{7}$$

This field is divergence-free for every t. The three-dimensional reference clouds use the same interacting particle method on $\mathbb{T}^3$. The first generation is uniform. The potential is $V_\lambda = \kappa\lambda^2 + \lambda v \cdot e_1$. Each generation again covers one forcing period with $\Delta t = 2^{-8}$ and a resample at the end of the period. After $T = 4096$ generations the $M = 2 \times 10^4$ terminal positions are stored at that σ. Training and evaluation reuse those stored clouds. The remaining discretization matches the two-dimensional experiment: $\lambda = 2$, training through $-\log_2 \sigma = 3.75$, and evaluation through 4.50. The eigenvalue metric is unchanged. The invariant-measure error is the W_2 distance of x-z projections,

$$\mathcal{E}_{xz}(\sigma) = W_2\big((\Pi_{xz})_{\#}\hat{\mu}^{(\sigma)}, (\Pi_{xz})_{\#}\hat{\nu}^{(\sigma)}\big), \tag{8}$$

where $\Pi_{xz}(x, y, z) = (x, z)$.

4.4.2 Projected invariant measure

At $\sigma = 2^{-4.5}$, the Feynman–Kac cloud is anisotropic across the three coordinate planes. Mass concentrates at larger z in the x-z and y-z views, and the y-z plane shows a curved ridge. This structure is consistent with a potential V_λ that depends on x_3 through the first velocity component. Figure 8 compares the three planes for the reference, Meanflow, and DP. Meanflow recovers the high-z concentration and the main anisotropy of each plane, with a noisier footprint than the reference. DP tightens those planes relative to Meanflow. The x-y view recovers the two opposing chevrons of the Feynman–Kac cloud, and the y-z ridge is thinner than the Meanflow band. Both generated clouds remain grainier than the reference. The x-z plane is the one used for the W_2 numbers below.

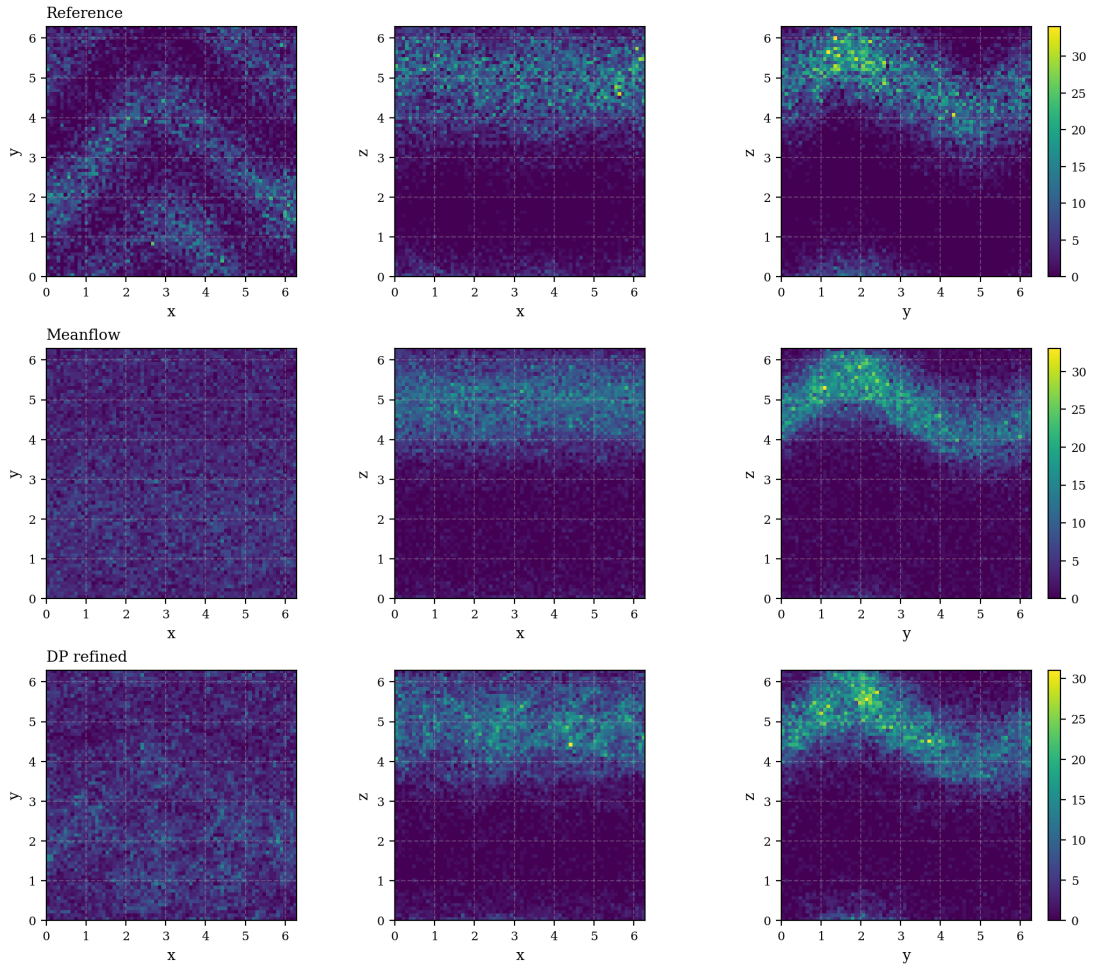


Figure 8: Three-dimensional invariant-measure projections at $\sigma = 2^{-4.5}$. Rows are the Feynman–Kac reference, Meanflow, and DP. Columns are the x-y, x-z, and y-z planes. Each row shares a colorbar.

4.4.3 Projected Wasserstein Error

The uniform cold start remains between 0.8895 and 0.9600 on the grid. Meanflow lowers the projected error to a mean of 0.1293, next to the source-paper Meanflow mean of 0.1312. DP has a grid mean of 0.1065 and a minimum of 0.0920 at $-\log_2 \sigma = 2.75$. The source-paper DP mean is 0.0294. At $-\log_2 \sigma = 4.50$, Meanflow is 0.1582 and DP is 0.1366. Figure 9 plots these two columns. DP is larger than Meanflow only at the coarsest training level and smaller at the remaining ten, including the farthest extrapolation point.

$-\log_2 \sigma$	Meanflow	DP refinement
2.00	0.0894	0.1047
2.25	0.1052	0.0997
2.50	0.1201	0.0954
2.75	0.1272	0.0920
3.00	0.1102	0.0988
3.25	0.1266	0.1016
3.50	0.1263	0.1126
3.75	0.1417	0.0945
4.00	0.1470	0.1168
4.25	0.1701	0.1191
4.50	0.1582	0.1366

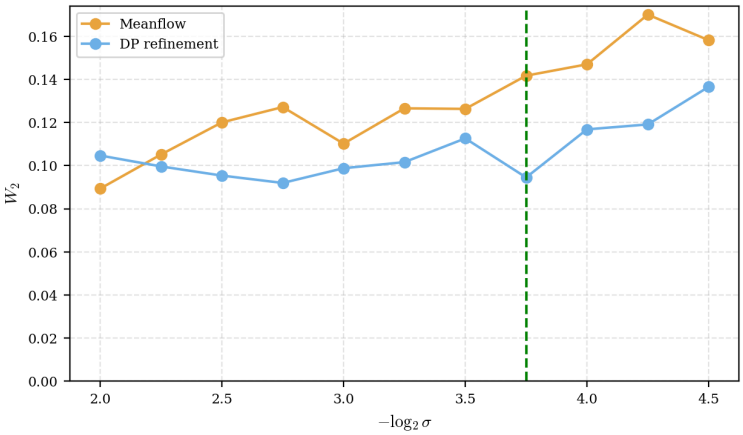


Figure 9: Projected x-z W_2 on $\mathbb{T}^3$. The left table and the right curves use the same Meanflow and DP values. The dashed vertical line is $-\log_2 \sigma = 3.75$.

4.4.4 Eigenvalue estimator

At $\sigma = 2^{-4.5}$, the long-horizon Feynman–Kac value is 0.6147. Figure 10 shows a cold start near 0.36 that reaches the reference by about 10^2 generations. Meanflow is 0.6148 at $T = 1$ and 0.6131 at $T = 128$. DP is 0.6171 at $T = 1$ and 0.6142 at $T = 128$. The source paper's DP warm start is 0.6135 at $T = 1$. As in two dimensions, the eigenvalue can sit near the Feynman–Kac limit while the projected W_2 remains of order 10^{-1}. The reproduced Meanflow projected W_2 sits near the source-paper Meanflow column. The DP refinement does not reach the source-paper DP table.

iteration	MF-only init	MF+DP init
1	0.614816	0.617057
2	0.608825	0.613084
4	0.600922	0.606998
8	0.599041	0.604442
16	0.606779	0.609265
32	0.611394	0.612179
64	0.612298	0.613662
128	0.613059	0.614218

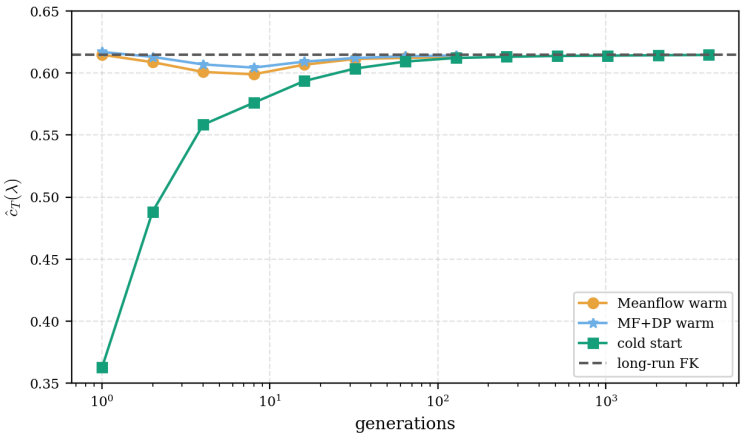


Figure 10: Three-dimensional front-speed estimator $\hat{c}_T(\lambda)$ at $\sigma = 2^{-4.5}$. The left table lists the two warm starts. The right panel adds the uniform cold start and the long-horizon Feynman–Kac reference (dashed). The horizontal axis is logarithmic in the number of generations.

References

[1] Z. Wang, J. Xin, and Z. Zhang. DeepParticle: learning invariant measure by a deep neural network minimizing Wasserstein distance on data generated from an interacting particle method. *J. Comput. Phys.* 464 (2022) 111309.